

Question				Marking details			Marks available																	
							AO1	AO2	AO3	Total	Maths	Prac												
5	(a)	(i)		D			1			1														
		(ii)		B			1			1														
		(iii)		E				1		1														
		(iv)	I	8 [V]				1		1	1													
			II	0.02 [s]				1		1	1													
			III	<table><tr><td>Change</td><td>Effect on maximum voltage</td><td>Effect on time for 1 rotation (spin)</td></tr><tr><td>Stronger magnets</td><td>increases</td><td>no change</td></tr><tr><td>More turns on the coil</td><td>increases (1)</td><td>no change</td></tr><tr><td>Coil spins slower</td><td>decreases (1)</td><td>increases (1)</td></tr></table>			Change	Effect on maximum voltage	Effect on time for 1 rotation (spin)	Stronger magnets	increases	no change	More turns on the coil	increases (1)	no change	Coil spins slower	decreases (1)	increases (1)		3		3		
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More turns on the coil	increases (1)	no change																						
Coil spins slower	decreases (1)	increases (1)																						
	(b)	(i)		Ticks in boxes 3, 5 and 6 i.e. The core is made from iron The primary coil creates an alternating magnetic field Transformers can increase or decrease voltages Deduct 1 mark for any additional tick			3			3														
		(ii)		Substitution: $20 \times \frac{36}{12}$ (1) $N_2 = 60$ (1)			1	1		2	2													
				Question 5 total			6	7	0	13	4	0												